

KOUSIK KUMAR DUTTA

Ph.D. in Computer Science & Engineering
Indian Institute of Technology Ropar

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About

Ph.D. researcher in Computer Science and Engineering at the Indian Institute of Technology Ropar, specializing in spatial data management, route planning, and scalable graph algorithms for large time-dependent networks. My research focuses on the design of efficient algorithms and systems for constrained optimization in dynamic transportation environments, encompassing preference-aware routing, interval-based optimization, and logistics-oriented decision making. Through a series of journal and conference publications, my work has progressively generalized constrained optimization formulations toward a unified framework that integrates both maximization and minimization objectives under diverse temporal and operational constraints. Looking ahead, I seek to advance scalable graph-processing systems and intelligent optimization techniques by combining algorithmic foundations, data systems, and high-performance computing methodologies.

Research Interests

- **Spatial Data Management:** Efficient storage, querying, and processing of large-scale spatio-temporal and road network data.
- **Route Planning and Network Optimization:** Preference-aware and constrained optimization methods for path computation in dynamic transportation networks.
- **Graph Algorithms for Large-Scale Networks:** Designing scalable algorithms for shortest-path and graph-search problems in large graphs.
- **Intelligent Transportation Systems:** Data-driven approaches for mobility analytics, navigation systems, and transportation optimization.

Education

Ph.D. in Computer Science and Engineering 2021–2026 *Indian Institute of Technology Ropar, Punjab, India*

- **Dissertation:** “Constrained Maximization of Preferences on Time-Dependent Road Networks”
- **Advisor:** Dr. Venkata M. V. Gunturi & Venkata Kalyan Tavva
- **CGPA:** 7.92/10.00

M.Tech in Computer Science and Engineering 2019–2021 *Indian Institute of Technology Ropar, Punjab, India*

- **Thesis:** “A Multi-Threading Algorithm for Constrained Path Optimization on Road Networks”
- **CGPA:** 8.39/10.00

B.Tech in Computer Science and Engineering 2014–2018 *Kalyani Government Engineering College, West Bengal, India*

- **CGPA:** 7.45/10.00

Publications

Journal Publications

- [J1] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. A Parallelizable Algorithm for Constrained Maximization of Preferences in Large Time-Dependent Graphs. *Geoinformatica*, vol. 30, article 19, 2026. DOI: <https://doi.org/10.1007/s10707-026-00577-z>

Conference Publications

- [C1] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. A User-Configurable Navigation System for Wideness and Turn-Aware Routing. *International Conference on Mobile Data Management (MDM)*, 2026. Accepted.
- [C2] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. Interval-Based Constrained Path Optimization in Time-Dependent Road Networks. *26th International Conference on Web Information Systems Engineering (WISE)*, 2025. DOI: https://doi.org/10.1007/978-981-95-7251-9_12
- [C3] **Kousik Kumar Dutta**, Ankita Dewan, Venkata M. V. Gunturi. A Multi-Threading Algorithm for Constrained Path Optimization on Road Networks. *23rd International Conference on Web Information Systems Engineering (WISE)*, 2022. DOI: https://doi.org/10.1007/978-3-031-20891-1_9
- [C4] **Kousik Kumar Dutta**, Prathamesh N. Tanksale, Shirshendu Das. A Fairness Conscious Cache Replacement Policy for Last Level Cache. *Design, Automation and Test in Europe (DATE)*, 2021. DOI: <https://doi.org/10.23919/DATE51398.2021.9474096>
- [C5] Ankita Dewan, Venkata M. V. Gunturi, Vinayak Naik, **Kousik Kumar Dutta**. NEAT Activity Detection Using Smartwatch at Low Sampling Frequency. *IEEE SmartWorld/SCALCOM/UIC/ATC/IOP/SCI*, 2021. DOI: <https://doi.org/10.1109/SWC50871.2021.00014>

Workshop Publications

- [W1] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. Constrained Path Optimization on Time-Dependent Road Networks. *WISE PhD Symposium, Demos and Workshops*, 2024. DOI: <https://doi.org/10.1007/978-981-96-1483-7>
- [W2] Prathamesh N. Tanksale, **Kousik Kumar Dutta**, Shirshendu Das, T. V. Kalyan. CAMOUFLAGE: An Efficient Mechanism to Hide Congestion in NVM LLC. Extended Abstract, *IEEE International Conference on High Performance Computing (HiPC)*, 2023. DOI: <https://doi.org/10.1109/HiPCW61695.2023.00021>

Manuscripts Under Review

- [U1] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. A Bi-Directional Search for Interval Constrained Path Optimization in Large Time-Dependent Road Networks. Submitted to *Knowledge and Information Systems*, 2026. (Under review)
- [U2] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. Streamlining Vehicle Logistics: Maximizing Pickup–Delivery under Non-Additive Loading–Unloading Costs. Submitted to *Journal of Scheduling*, 2026. (Under review)

Teaching Experience

Teaching Assistant, Indian Institute of Technology Ropar

- **Databases (B.Tech. Core Course)** 7 Offerings (2021–2025)
 - Conducted laboratory sessions and evaluated assignments on SQL, relational database design, and query processing.
 - Mentored student teams in the development of database-driven applications and course projects.
- **Software Engineering (B.Tech. Core Course)** 5 Offerings (2020–2025)
 - Evaluated laboratory assignments and guided students through software development practices.
 - Assisted student teams in requirement analysis, software design, testing, and project implementation.
- **PG Software Lab (M.Tech. Core Course)** 1 Offering (2020)
 - Mentored postgraduate students in advanced data structures and algorithm implementation.

Academic Services

- **Reviewer** for *Geoinformatica* (2023–Present); completed peer review of three manuscripts in areas including spatial databases, graph processing, and route planning.
- **Technical Committee Head**, *SYNTACS 2026* (Symposium on Novel Technologies and Advances in Computer Science), Department of Computer Science and Engineering, Indian Institute of Technology Ropar. Led the technical committee and supervised technical activities including poster organization and evaluation, while coordinating the overall execution and management of technical components of the symposium.

Awards, Fellowships, and Grants

- **IEEE TCDE Student Grant (2026)**: Recipient of the IEEE Technical Committee on Data Engineering (TCDE) Student Grant, awarded competitively to support participation in the International Conference on Mobile Data Management (MDM 2026).
- **Institute Fellowship (Ph.D.)**: Awarded full institute fellowship for doctoral studies at Indian Institute of Technology Ropar (2021–2026).
- **MHRD Scholarship (M.Tech.)**: Awarded Ministry of Human Resource Development (MHRD) scholarship during M.Tech studies at Indian Institute of Technology Ropar (2019–2021).
- **GATE 2019**: Secured All India Rank (AIR) 644 in Computer Science and Engineering.

References

1. **Dr. Venkata M. V. Gunturi** Email: V.Gunturi@lboro.ac.uk
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Loughborough University, United Kingdom
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Web: <https://www.lboro.ac.uk/departments/compsci/staff/venkata-maruti-viswanath-gunturi/>
2. **Dr. Shirshendu Das** Email: shirshendu@cse.iith.ac.in
Associate Professor, Department of Computer Science and Engineering
Indian Institute of Technology Hyderabad, India
Relationship: Research Collaborator / M.Tech Mentor
Web: <https://sites.google.com/view/shirshendudas>
3. **Dr. Manish Kumar** Email: manishk@iitrpr.ac.in
Assistant Professor, Department of Computer Science and Engineering
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Relationship: Teaching Assistant Instructor
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